

Shubhangi Agarwal

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SUMMARY

I am a researcher with experience in **data mining**, **machine learning**, and **large-scale graph data** analysis. Previously, I worked on a generative AI initiative for Indian languages in collaboration with IIT Kanpur, where I worked extensively with domain experts and linguists to develop an end-to-end LLM-based data generation pipeline using **prompt engineering**. I developed statistical and algorithmic methods for subgraph querying in complex networks, with relevance to problems such as information extraction, recommendation systems, and fraud detection. My postdoctoral work focused on HyGraphs, a unified data model for dynamic graph and time-series data, enabling efficient analysis of evolving large-scale datasets.

EDUCATION

Ph.D. in Computer Science and Engineering

Indian Institute of Technology Kanpur, Uttar Pradesh, India

CGPA: 8.25

2014 - 2023

Bachelor of Technology Computer Engineering

Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India

CGPA: 8.71

2010 - 2014

PHD THESIS

Subgraph Matching and Mining in Large Graphs

Supervisor: Arnab Bhattacharya

- Developed algorithms for Approximate Subgraph Matching in both deterministic and probabilistic graphs.
- Proposed a Graph Neural Network model for robust node embeddings with positional information.

PUBLICATIONS

- “FogCast: Probabilistic Medium Range Fog Forecasting using Global Machine Learning Weather Prediction Model”, Prasad Deshpande, **Shubhangi Agarwal**, Kritika Bansal, Arnab Bhattacharya and Shivam Tripathi, Quarterly Journal of the Royal Meteorological Society, 152(777), e70150.
- “A Case Study of Cross-Lingual Zero-Shot Generalization for Classical Languages in LLMs”, V S D S Mahesh Akavarapu, Hrishikesh Terdalkar, Primit Bhattacharyya, **Shubhangi Agarwal**, Vishakha Deulgaonkar, Chaitali Dangarikar, Pralay Manna, and Arnab Bhattacharya, 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025, Austria.
- “Towards Hybrid Graphs: Unifying Property Graphs and Time Series”, Mouna Ammar, Christopher Rost, Riccardo Tommasini, **Shubhangi Agarwal**, Angela Bonifati, Petra Selmer, Evgeny Kharlamov and Erhard Rahm, 28th International Conference on Extending Database Technology (EDBT), 2025, Spain.
- “Transforming Time Series into Graphs and Back with HyGraph”, Mouna Ammar, **Shubhangi Agarwal**, Angela Bonifati and Erhard Rahm, Proceedings of the Workshops of the EDBT/ICDT Joint Conference, 2025, Spain.
- “VeNoM: Approximate Subgraph Matching with Enhanced Neighbourhood Structural Information”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, 7th Joint International Conference on Data Science and Management of Data (CODS-COMAD), 2024, India.
- “VerSaChI: Finding Statistically Significant Subgraph Matches using Chebyshev’s Inequality”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Information and Knowledge Management (CIKM), 2021, pages 2812-2816, Australia.
- “GraphReach: Position-Aware Graph Neural Network using Reachability Estimations”, Sunil Nishad, **Shubhangi Agarwal**, Arnab Bhattacharya and Sayan Ranu, Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), 2021, pages 1527-1533, Canada.
- “ChiSeL: Graph Similarity Search using Chi-Squared Statistics in Large Probabilistic Graphs”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Very Large Data Bases (VLDB), 2020, pages 1654-1668, Japan.

EXPERIENCE

Software Engineer

Google IT Services India Private Limited, Bengaluru, India

Aug 2026 - Present

Project Executive Officer

IIT Kanpur, India

Sep 2025 - Jul 2026

- Project: 'BharatGPT: A Suite of Generative AI Tech for India'.
- Developed an end-to-end pipeline for dataset creation, from generating and processing LLM outputs to evaluating them against expert-annotated data, using prompts developed in collaboration with domain experts and linguists to create a feature-rich language representation.
- Funding Agency: Department of Science and Technology (DST), Government of India.

Post-doctoral Researcher

LIRIS Research Lab, CNRS, Université Claude Bernard Lyon 1, France

May 2024 - Apr 2025

- Project: 'HyGraph: Querying and Analytics on Hybrid Graphs'.
- Conducted preliminary research on hybrid graph and time-series storage solutions to support evolving data needs.
- Funding Agency: French Agence Nationale de la Recherche (ANR) and the Deutsche Forschungsgemeinschaft (DFG; German Research Foundation).

Senior Student Research Associate

IIT Kanpur, India

Aug 2022 - Dec 2023

- Project: 'Short and Long-term Fog Prediction for North Indian Cities'.
- Conducted preliminary research on hybrid graph and time-series storage solutions to support evolving data needs.
- Funding Agency: Scheme for Transformational and Advanced Research in Science (STAR), Ministry of Education, Government of India.

Summer Intern

Flipkart Internet Private Limited, Bengaluru, India

Jun 2015 - Jul 2015

- Worked in a project to prepare a model for identification of 'fake' listings.

Summer Research Internship

IIT Gandhinagar, India

May 2012 - July 2012

- Worked under Prof. Pawan Lingras on Time Series Analysis

TEACHING

Instructor

Université Claude Bernard Lyon 1, France

Sep 2024 - Dec 2024

- Program: M2 - Data and Intelligence for Smart Systems (DISS)
- Course: Data Processing and Analytics (Jointly with others)

Teaching Assistant

IIT Kanpur, India

Aug 2014 - Apr 2021

- Graded and evaluated projects for various courses of Computer Science.

Senior Tutor

IIT Kanpur, India

Aug 2017 - Apr 2018, Aug 2019 - Apr 2020

- Led teams of strength ~60; Assisted in backend management, paper-setting and grading.

Teaching Assistant for MOOC

NPTEL - Remote

July 2017 - Sep 2017

- Crafted objective questions and resolved student queries on Fundamentals of Database Systems.

ADMINISTRATIVE EXPERIENCES

Program Committee Member

2025 - current

- GRADES-NDA Workshop - SIGMOD/ PODS (2025), ICWE (2025, 2026), TGD Workshop - EDBT/ ICDT (2025).

External Reviewer

2020 - current

- WSDM (2024), CIKM (2021, 2022), DASFAA (2022), CoDS-COMAD (2020, 2021), KDD (2021).

TECHNICAL SKILLS

Languages : C, C++, Python, Java, R, JavaScript, PHP, Shell scripting
Databases : MySQL, Neo4j, SQLite
Libraries : PyTorch, Tensorflow, Scikit-learn, Numpy, Pandas, PySpark, Ollama
Tools : git, LaTeX, Docker, Weka

REFERENCES

Arnab Bhattacharya

Professor, Indian Institute of Technology Kanpur, India

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Sourav Dutta

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Angela Bonifati

Distinguished Professor, Lyon 1 University, France

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Riccardo Tommasini

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